
Enforcing Spectrally Consistent Diffusion-Based Cloud Removal in Remote Sensing Images

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Abstract

Cloud cover remains a pervasive issue in optical remote sensing. Recent advances in generative modeling, particularly in diffusion models, have established a new state-of-the-art in cloud removal. However, existing approaches struggle with explicitly enforcing spectral consistency, or the requirement that individual spectral bands retain their physically meaningful relationships and inter-band correlations, which can introduce spectral anomalies and artifacts that undermine the integrity of subsequent analyses. To address this limitation, we propose a framework that integrates state-of-the-art cloud removal diffusion model architectures with Neural Approximate Mirror Maps (NAMMs), a class of constraint-satisfaction operators capable of enforcing general, possibly non-convex constraints during the generative process. We incorporate a constraint loss to guide the sampling procedure, resulting in cloud-free reconstructions that are both visually coherent and spectrally consistent. Our implementation is available at <https://github.com/weikaylee/cs159>.

1 Introduction

Cloud removal is an important problem in optical remote sensing because clouds often obscure the land surface and limit the usefulness of satellite imagery. Recent diffusion-based methods have shown strong performance for reconstructing cloud-free images [1]. However, in multispectral remote sensing, visual quality alone is not enough: each spectral band carries physical information about the scene, and the relationships between bands are important for tasks such as vegetation analysis, land-cover classification, and environmental monitoring [2]. A cloud-free reconstruction can therefore look visually plausible while still distorting the spectral structure of the original image.

This creates a key challenge for diffusion-based cloud removal: standard diffusion models do not explicitly guarantee that the generated image preserves physically meaningful spectral relationships. This limitation is especially important for Sentinel-2 imagery, since the SEN12MS-CR dataset [2] contains 13-band multispectral Sentinel-2 images. Existing cloud removal models such as EMRDM [1] achieve strong reconstruction performance by formulating cloud removal as a mean-reverting denoising process conditioned on cloudy inputs. However, these models still do not directly enforce spectral consistency as a constraint during generation.

In this project, we investigate whether Neural Approximate Mirror Maps (NAMMs) can improve spectral consistency in diffusion based cloud removal. NAMMs are designed for constrained generative modeling: they learn a forward map that transforms constrained data into an unconstrained

mirror space and an inverse map that brings generated samples back toward the constraint set. This framework is useful for our project because spectral consistency can be expressed through differentiable loss terms rather than a closed form constraint. In this work, we define spectral consistency as the preservation of (1) per-pixel spectral signatures and inter-band relationships, measured through Spectral Angle Mapper (SAM), and (2) global per-band reflectance statistics, measured through moment matching. This definition provides a practical and differentiable objective for enforcing physically meaningful spectral properties during generation. Together, these components encourage generated cloud-free images to preserve local spatial structure, spectral shape, and global reflectance statistics.

We evaluate our proposed approach on the SEN12MS-CR by training the NAMM directly on the cloud free images, then training an EDM-style diffusion model in the learned mirror space. During inference, generated mirror-space samples were mapped back into the constrained image space through the inverse NAMM. This allowed us to test whether the mirror space based approach helps produce reconstructions that remain both visually and spectrally consistent. Our results suggest that NAMM-based constrained generation can preserve strong reconstruction quality while keeping MAE comparable and spectral performance, demonstrating that mirror-space diffusion is a promising direction for physically constrained remote sensing reconstruction.

2 Background

2.1 Elucidated Diffusion Models

Elucidated Diffusion Models (EDMs), introduced by Karras et al. [3], provide a framework for diffusion-based generative modeling that improves training stability and sample quality through carefully designed noise parameterizations and network preconditioning. Unlike score-based diffusion methods that learn the gradient of the data distribution and require solving a reverse-time stochastic differential equation during sampling, EDMs train a denoising network that directly predicts clean samples from noisy inputs.

Given a clean sample x and noise level σ , EDM constructs a noisy observation

$$x_\sigma = x + \sigma\epsilon$$

where $\epsilon \sim \mathcal{N}(0, I)$. The model is parameterized as a preconditioned denoiser:

$$D_\theta(x_\sigma, \sigma) = c_{skip}(\sigma)x_\sigma + c_{out}(\sigma)F_\theta(c_{in}(\sigma)x_\sigma, \sigma)$$

where F_θ is a neural network and c_{in} , c_{out} , c_{skip} are deterministic functions of σ and the data standard deviation σ_{data} .

In this work, EDM serves as the generative model operating in the learned mirror space. After training the forward and inverse mirror maps described in Section 2.2, cloud-free Sentinel-2 images are mapped into mirror space and used to train a conditional EDM. The diffusion model is conditioned on Sentinel-1 SAR measurements and cloudy Sentinel-2 observations, and generation is performed entirely within the unconstrained mirror-space representation. The resulting samples are subsequently mapped back to image space through the learned inverse mirror map, producing spectrally consistent cloud-free reconstructions.

2.2 Neural Approximate Mirror Maps

Feng et al. [4] propose Neural Approximate Mirror Maps (NAMMs) to enable diffusion models to generate samples satisfying general, possibly non-convex constraints. The core idea is to learn a forward map g_ϕ that sends the constrained set $\mathcal{M} \subset \mathbb{R}^d$ to an unconstrained "mirror" space, and an approximate inverse f_ψ that maps back. A standard diffusion model is trained in the unconstrained mirror space; at inference, generated samples are projected back to $\mathcal{M}$ via f_ψ .

Both maps are trained jointly by minimizing

$$\phi^*, \psi^* \in \arg \min_{\phi, \psi} \{ \mathcal{L}_{cycle}(\mathbf{g}_\phi, \mathbf{f}_\psi) + \lambda_{constr} \mathcal{L}_{constr}(\mathbf{g}_\phi, \mathbf{f}_\psi) + \lambda_{reg} \mathcal{R}(\mathbf{g}_\phi) \}, \quad (1)$$

where $\mathcal{L}_{cycle}$ enforces mutual invertibility of g and f , $\mathcal{L}_{constr}$ penalizes constraint violations of reconstructed points, and $\mathcal{R}$ is a regularizer promoting a unique solution. The constraint loss averages

over noise levels up to $\sigma_{\max}$ to ensure robustness of f_ψ to diffusion model sampling error:

$$\mathcal{L}_{\text{constr}}(\mathbf{g}_\phi, \mathbf{f}_\psi) := \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[\int_0^{\sigma_{\max}} \mathbb{E}_{\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [\ell_{\text{constr}}(\mathbf{f}_\psi(\mathbf{g}_\phi(\mathbf{x}) + \sigma \mathbf{z}))] d\sigma \right], \quad (2)$$

where ℓ_{constr} is a differentiable distance to the constraint set. The forward map g_ϕ is parameterized as the gradient of an input-convex neural network (ensuring injectivity), and f_ψ as a ResNet. After joint training, an optional fine-tuning stage adapts f_ψ to reduce distribution shift between the Gaussian training noise and the actual sampling errors of the diffusion model.

To follow this framework, we define ℓ_{constr} to enforce spectral consistency, as defined in Section 3, and train an EDM model in the mirror space.

3 Methods

We investigate whether using the NAMM [4] framework improves cloud removal performance on the SEN12MS-CR dataset [2]. Specifically, we ask: can a learned constrained diffusion model better preserve the spectral properties of multispectral satellite imagery compared to an unconstrained baseline?

We define a differentiable constraint distance function ℓ_{constr} as a combination of three complementary terms operating at different levels of spectral and spatial fidelity:

$$\ell_{\text{constr}}(\hat{\mathbf{x}}, \mathbf{x}) = \mathcal{L}_{\text{recon}}(\hat{\mathbf{x}}, \mathbf{x}) + \lambda_{\text{SAM}} \mathcal{L}_{\text{SAM}}(\hat{\mathbf{x}}, \mathbf{x}) + \lambda_{\text{mom}} \mathcal{L}_{\text{moments}}(\hat{\mathbf{x}}, \mathbf{x}), \quad (3)$$

where $\hat{\mathbf{x}} = f_\psi(\mathbf{g}_\phi(\mathbf{x}) + \sigma \mathbf{z})$ is the image recovered by the inverse map from a noisy mirror-space point, and $\mathbf{x}$ is the cloud-free reference. The three terms enforce local spatial fidelity, spectral shape consistency, and global reflectance statistics, encouraging images that are physically consistent with Sentinel-2 observations. NAMM’s construction provides a guarantee that final samples satisfy the constraint set.

3.1 Reconstruction loss.

The first term is a pixel-wise mean squared error:

$$\mathcal{L}_{\text{recon}}(\hat{\mathbf{x}}, \mathbf{x}) = \|\hat{\mathbf{x}} - \mathbf{x}\|_2^2, \quad (4)$$

which provides dense, spatially localized supervision and forms the primary signal driving f_ψ to invert g_ϕ accurately.

3.2 Spectral Angle Mapper loss

The second term measures the mean per-pixel spectral angle between $\hat{\mathbf{x}}$ and $\mathbf{x}$ [5]:

$$\mathcal{L}_{\text{SAM}}(\hat{\mathbf{x}}, \mathbf{x}) = \frac{1}{N} \sum_{n=1}^N \arccos \left(\frac{\hat{\mathbf{x}}_n^\top \mathbf{x}_n}{\|\hat{\mathbf{x}}_n\| \|\mathbf{x}_n\|} \right), \quad (5)$$

where $\hat{\mathbf{x}}_n, \mathbf{x}_n \in \mathbb{R}^C$ are the spectral vectors at pixel n and $N = BHW$ is the total number of pixels. Because $\mathcal{L}_{\text{SAM}}$ is scale-invariant, it is insensitive to global brightness differences and penalizes only distortions in the shape of the spectrum, or the relative ratios between bands that determine physically meaningful quantities such as vegetation indices and land-cover class. This is a constraint that $\mathcal{L}_{\text{recon}}$ alone cannot enforce, as two images may have low MSE while exhibiting significant band-ratio corruption.

3.3 Moment matching loss

The third term enforces agreement of the per-band mean and standard deviation between $\hat{\mathbf{x}}$ and $\mathbf{x}$:

$$\mathcal{L}_{\text{moments}}(\hat{\mathbf{x}}, \mathbf{x}) = \|\boldsymbol{\mu}^{\hat{\mathbf{x}}} - \boldsymbol{\mu}^{\mathbf{x}}\|_2^2 + \|\boldsymbol{\sigma}^{\hat{\mathbf{x}}} - \boldsymbol{\sigma}^{\mathbf{x}}\|_2^2, \quad (6)$$

where $\boldsymbol{\mu}, \boldsymbol{\sigma} \in \mathbb{R}^C$ are the vectors of per-band means and standard deviations, averaged over batch and spatial dimensions. While $\mathcal{L}_{\text{recon}}$ enforces local pixel-level fidelity, $\mathcal{L}_{\text{moments}}$ enforces global reflectance consistency.

We evaluate our model on the SEN12MS-CR [2] benchmark using standard MAE, SAM, PSNR, and SSIM metrics.

4 Experiments

4.1 SEN12MS-CR Dataset

All experiments are conducted on the SEN12MS-CR dataset [2], a global multi-modal remote sensing dataset specifically developed for cloud removal in satellite imagery. Each sample in the dataset is a triple consisting of (1) 2-band Sentinel-1 SAR measurements, (2) 13-band cloud-covered Sentinel-2 imagery, and (3) 13-band cloud-free Sentinel-2 imagery. The dataset contains more than 122,000 image patch triplets collected from 175 globally distributed regions that cover diverse biomes, seasons, and atmospheric conditions. A key advantage of the SEN12MS-CR dataset is the multimodal design, as the Sentinel-1 SAR measurements are largely unaffected by atmospheric conditions. These SAR measurements can assist in recovering the missing optical information in the cloud-covered images. This dataset has become a widely adopted benchmark in remote sensing image restoration and cloud removal research, providing a standardized basis for evaluating model performance against numerous existing cloud-removal approaches, including recent diffusion-based methods.

4.2 Performance Metrics

Performance is evaluated using four quantitative metrics: Mean Absolute Error (MAE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Spectral Angle Mapper (SAM). MAE, PSNR, and SSIM assess image reconstruction quality by measuring pixel-wise error, signal fidelity, and structural similarity between the reconstructed image and the ground-truth cloud-free image. SAM measures the angular difference between spectral vectors, providing a direct measure of spectral consistency, and is therefore the primary metric in evaluating the effectiveness of the proposed approach. The SEN12MS-CR dataset reports benchmark results using these same metrics, enabling direct comparison with prior models.

4.3 Experimental Setup

We evaluate whether enforcing spectral consistency through Neural Approximate Mirror Maps (NAMMs) improves diffusion-based cloud removal on the SEN12MS-CR benchmark. Our primary baseline is an unconstrained EDM model trained directly in image space. The proposed approach first learns a NAMM consisting of a forward mirror map g_ϕ and inverse map f_ψ , then trains an identical EDM architecture in the learned mirror space. During inference, generated mirror-space samples are mapped back to the image domain using the learned inverse map.

To determine an effective mirror-space representation, we evaluated three candidate NAMM configurations that varied in the architecture and parameterization of the forward and inverse maps. The best-performing configuration was subsequently used for all downstream experiments.

The NAMM was trained using the constraint objective described in Section 3, combining reconstruction loss, Spectral Angle Mapper (SAM) loss, and moment matching loss. The mirror-space EDM was then trained on the transformed representations produced by the learned forward map. A batch size of 16 and a learning rate of $2e-4$ was used to train the model. The NAMM mirror map was trained for 80 epochs, while the EDM model was trained for 100 epochs.

For comparison, we train an EDM baseline using the same architecture and settings as the proposed model, but without applying the NAMM transformation. This baseline operates directly in the original Sentinel-2 image space and serves to isolate the effect of the learned mirror-space representation on reconstruction quality and spectral consistency.

For all experiments, performance was evaluated using the MAE, PSNR, SSIM, and SAM metrics on the SEN12MS-CR test set. Lower MAE and SAM indicate better reconstruction fidelity and spectral consistency, while higher PSNR and SSIM indicate improved image quality.

4.4 Mirror Map Architecture Selection

Before training the final diffusion model, we conducted an ablation study to determine an effective architecture and loss configuration for the NAMM mirror maps. In particular, we investigated how the relative weighting of spectral and statistical constraints affects the quality of the learned mirror space and its suitability for downstream diffusion modeling.

We varied the weighting of the Spectral Angle Mapper (SAM) loss and the moment matching losses defined in Section 3.2 and 3.3 across the different configurations. Table 1 summarizes the specific settings.

Table 1: NAMM ablation study configurations varying the weights of SAM and moment matching losses.

Configuration	λ_{SAM}	λ_{mom}
Configuration 1	0.1	1
Configuration 2	1	10
Configuration 3	10	10

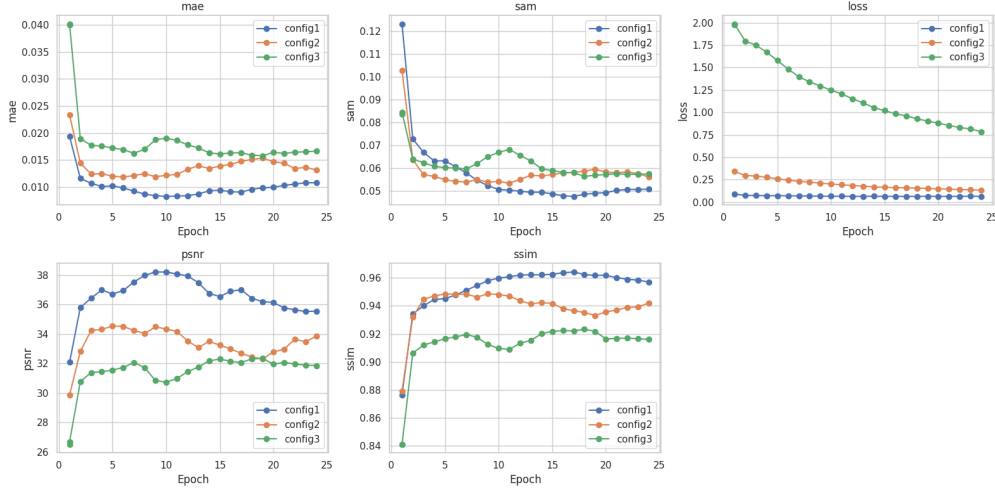


Figure 1: **Training curves for the three NAMM configurations.** The configuration settings are specified in Table 1. For each configuration, we evaluate the reconstruction quality using the MAE, PSNR, SSIM, and SAM metrics, as well as the total loss.

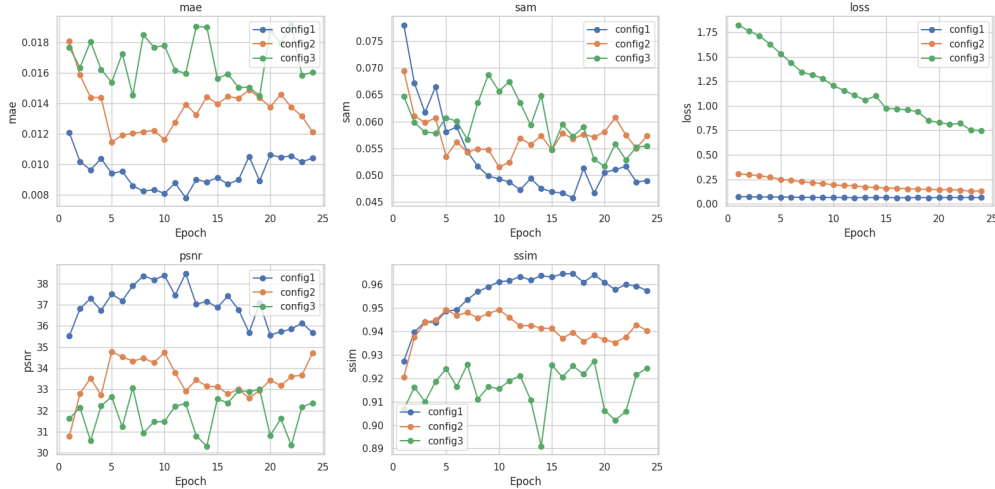


Figure 2: **Validation curves for the three NAMM configurations.** The configuration settings are specified in Table 1. For each configuration, we evaluate the reconstruction quality using the MAE, PSNR, SSIM, and SAM metrics, as well as the total loss.

Figures 1 and 2 show the training and validation curves, respectively, for all three configurations. Across all metrics, configuration 1 with $\lambda_{\text{SAM}} = 0.1$ and $\lambda_{\text{mom}} = 1$ achieved the best results as well as the lowest loss, outperforming the remaining configurations. Increasing the weights of the SAM and moment-matching losses beyond Configuration 1 consistently degraded both training and validation performance. This suggests that excessively strong spectral regularization can interfere with accurate reconstruction, indicating a trade-off between enforcing spectral constraints and preserving image fidelity. Consequently, configuration 1 was selected for all subsequent experiments.

4.5 Quantitative Results

Table 2 compares the proposed NAMM-constrained EDM against the unconstrained EDM baseline, as well as existing cloud-removal methods reported on the SEN12MS-CR benchmark.

Table 2: Comparison of cloud removal methods on the test dataset. Our results are highlighted in bold.

Method	MAE ↓	SAM ↓	PSNR ↑	SSIM ↑
EDM	0.104	35.896	18.58	0.536
McGAN	0.048	15.676	25.14	0.744
SAR-Opt-cGAN	0.043	15.494	25.59	0.764
SAR2OPT	0.042	14.788	25.87	0.793
SpA GAN	0.045	18.085	24.78	0.754
Simulation-Fusion GAN	0.045	16.633	24.73	0.701
DSen2-CR	0.031	9.472	27.76	0.874
GLF-CR	0.028	8.981	28.64	0.885
UnCRtainTS L2	0.027	8.320	28.90	0.880
ACA-Net	0.025	7.770	29.78	0.896
DiffCR	0.019	5.821	31.77	0.902
EMRDM	0.018	5.267	32.14	0.924
NAMM	0.019	5.008	32.43	0.892

Our method achieves an MAE of 0.019, PSNR of 32.43, SSIM of 0.892, and SAM of 5.008 on the test set. The unconstrained EDM baseline achieves an MAE of 0.104, PSNR of 18.58, SSIM of 0.536, and SAM of 35.896 on the test set. The unconstrained EDM baseline isolates the contribution of the NAMM framework, allowing us to evaluate whether enforcing spectral constraints through a learned mirror-space representation improves reconstruction quality beyond what can be achieved by diffusion alone. Compared with the unconstrained EDM baseline, the NAMM-based approach performs substantially better on all metrics. These improvements suggest that a mirror-space representation guided by spectral consistency constraints can improve reconstruction quality while preserving physically meaningful spectral relationships.

Compared with previously reported methods on SEN12MS-CR, our approach remains competitive across all evaluation metrics. Notably, the proposed method achieves the lowest SAM score among all methods reported in Table 2, indicating the strongest spectral consistency while maintaining reconstruction quality comparable to state-of-the-art diffusion-based approaches. These results suggest that learning a constrained mirror-space representation allows diffusion models to satisfy spectral objectives without significantly degrading image fidelity.

Overall, the results support the central hypothesis of this work: explicitly enforcing spectral consistency through NAMMs can improve diffusion-based cloud removal while preserving the advantages of modern generative reconstruction methods.

4.6 Qualitative Results

Figure 3 shows representative mirror-space EDM predictions. The generated mirror-space representations preserve meaningful scene structure and can be successfully mapped back to realistic cloud-free images through the inverse NAMM. This suggests that the learned mirror space captures a representation that encodes the desired spectral constraints while remaining suitable for diffusion-based generation.

Figure 4 visualizes the residuals produced by the NAMM reconstruction process. The residual maps highlight the regions where the inverse mirror map applies corrections to enforce consistency with the learned constraint set. The largest corrections are concentrated in areas affected by cloud contamination and complex land-cover transitions, indicating that the NAMM primarily acts where reconstruction errors are most likely to violate spectral constraints.

Together, these visualizations provide evidence that the proposed framework is not merely memorizing cloud-free imagery but instead learns a structured transformation that guides generation toward spectrally consistent solutions. The qualitative results are consistent with the quantitative improvements reported in Table 2 and demonstrate the effectiveness of constrained diffusion in multi-spectral remote sensing applications.

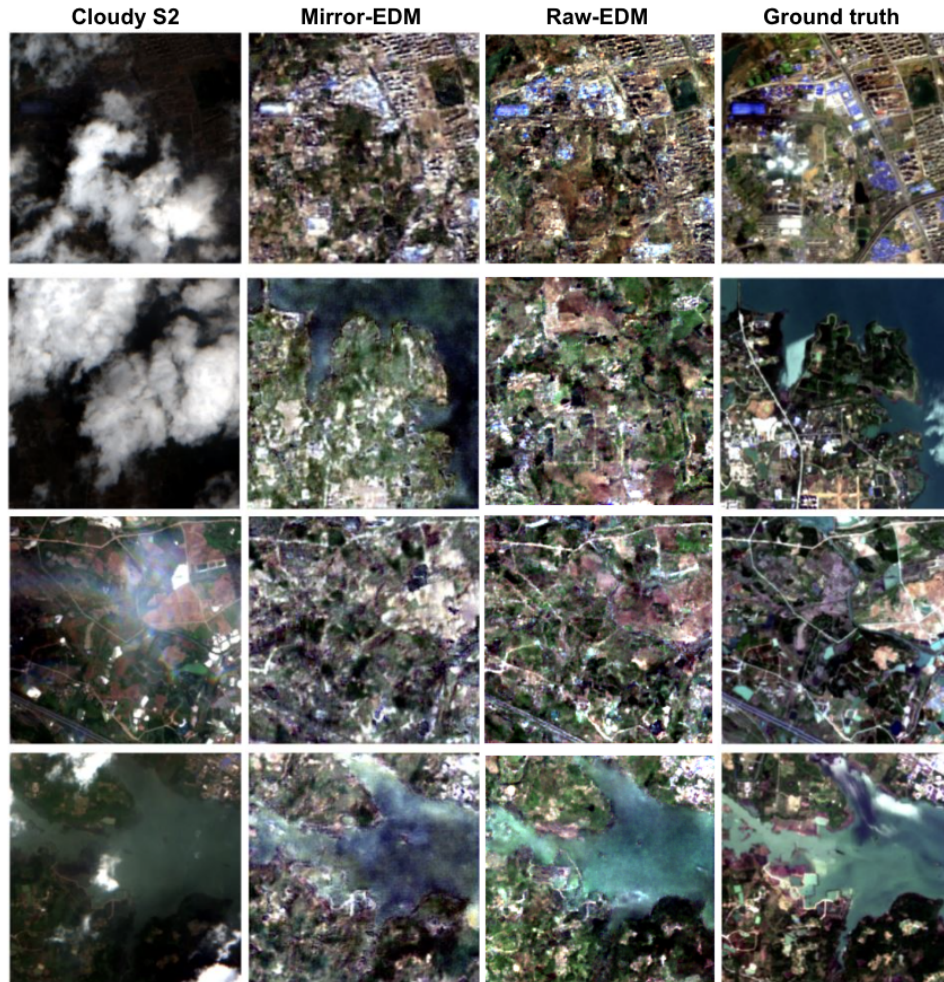


Figure 3: **Mirror-space vs raw baseline EDM predictions.** Cloud-free generation with the NAMM-EDM model with 0.15 dropout for cloud conditioning and the corresponding baseline-EDM model. Qualitatively, the baseline matches the ground truth aesthetically, but the mirror-EDM matches spectrally, as emphasized in the second row.

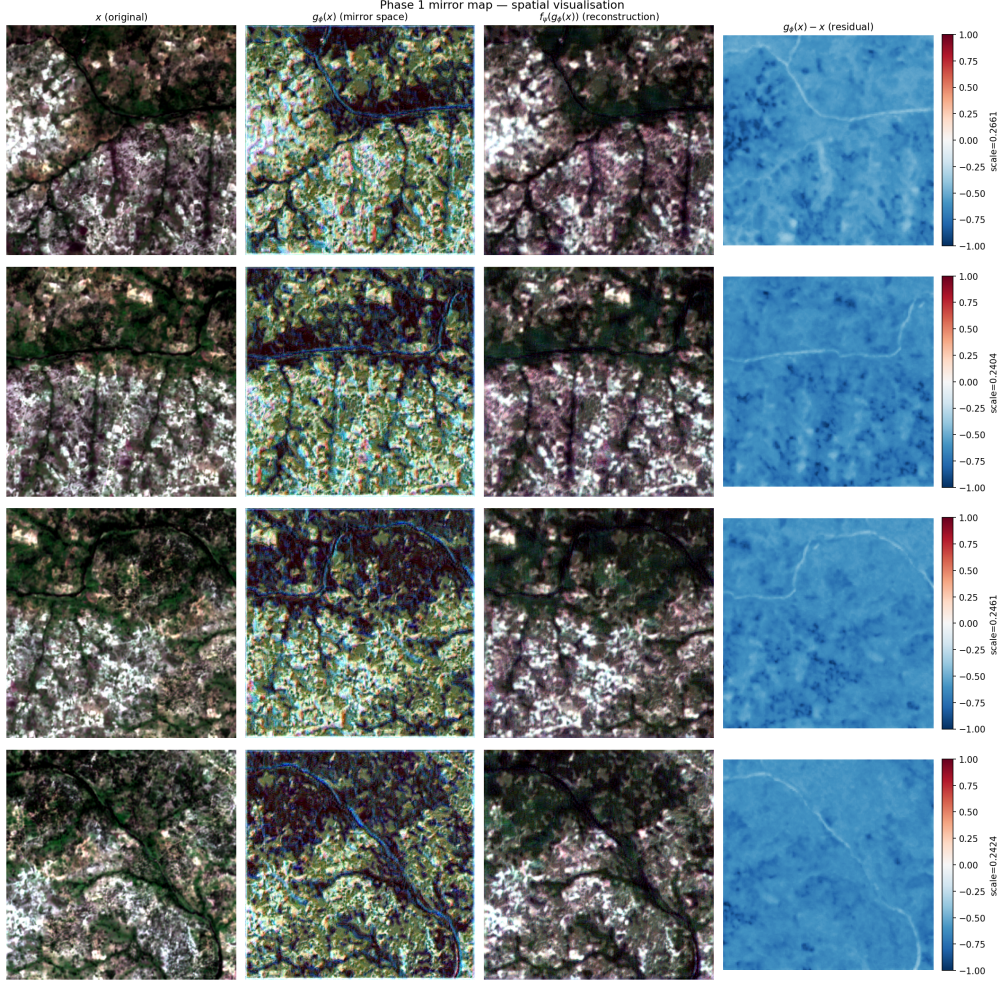


Figure 4: **Mirror map reconstructions.** The first column represents the original cloud-free ground truth, the second column represents the mirror space representation, and the third column represents the reconstruction of the ground truth using the reverse mapping. The last column shows the residual between the original input and the reconstructed input. Notably, all residuals are extremely low. [4]

5 Discussion

This project investigated whether Neural Approximate Mirror Maps (NAMMs) can be used to enforce spectral consistency in diffusion based cloud removal. The motivation was that cloud removal is not only a visual reconstruction problem. Instead, for multispectral satellite imagery, the generated cloud-free images should also preserve meaningful relationships between spectral bands. Standard diffusion models can produce visually consistent outputs, but it is not guaranteed that the reconstructed images satisfy the spectral constraints. By combining a mirror-space generative model with a constrained loss based on reconstruction error, Spectral Angle Mapper loss, and moment matching, our goal was to reach reconstructions that are both visually and spectrally consistent.

Our results show that NAMM based constrained generation is a promising direction for cloud removal, but also that the integration is not straightforward. The final NAMM-based model achieved an MAE of 0.019, which is close to the EMRDM’s baseline MAE value of 0.018. The model also achieved a PSNR of 32.43 dB, which is slightly higher than EMRDM’s PSNR of 32.14 dB. This suggests that the mirror-space approach can maintain reconstruction quality while adding an explicit spectral constraint.

The strongest improvement appears in the spectral metric. The NAMM model achieved a SAM score of 5.008 compared to EMRDM’s 5.267. Since SAM directly measures spectral consistency, this suggests that the NAMM constraint helped preserve spectral relationships between the bands. However, the NAMM model had a lower SSIM of 0.892 compared to EMRDM’s 0.924, which suggests that some structural detail may have been lost during the mirror-space transformation and inverse mapping process.

We also discovered several challenges during the experiments. Our first approach attempted to train the NAMM using EMRDM generated images, but this led to unstable training. In the second cycle, we trained the NAMM directly on SEN12MS-CR cloud-free images, and then we trained an EDM-style diffusion model in the mirror space. This pipeline was more stable and became our final approach.

Overall, these results show that NAMMs provide a strong foundation for incorporating differentiable constraints into cloud removal. At the same time, the method doesn’t yet clearly outperform the strongest baseline across all metrics. These results suggest that mirror-space-based diffusion is a promising approach for spectrally constrained image generation. Although the proposed method achieves competitive reconstruction performance and improved spectral fidelity, additional refinement is needed before it can consistently outperform the strongest existing cloud-removal methods across all evaluation metrics.

6 Future Work

A natural extension of this work is to integrate the Neural Approximate Mirror Maps (NAMMs) framework with the Effective Mean-Reverting Denoising Model (EMRDM) [1], which currently represents the strongest diffusion-based cloud-removal architecture on the SEN12MS-CR dataset. Initial experiments following this approach revealed a fundamental incompatibility between EMRDM’s mean-reverting diffusion formulation and the learned mirror-space representation. EMRDM constructs a mean-reverting residual diffusion bridge between cloudy and cloud-free observations. However, NAMM first transforms the target distribution into the learned mirror space. As a result, the diffusion process attempts to interpolate between cloudy Sentinel-2 observations in image space and targets in mirror space. Because these representations lie in different domains and exhibit substantially different value ranges and distributions, the model consistently collapsed toward near-zero outputs, leading to unstable optimization.

Future work could investigate alternative formulations that preserve EMRDM’s conditioning mechanism while constructing the diffusion process entirely within mirror space. One possibility is to learn mirror-space representations for both the conditioning inputs and reconstruction targets, enabling a spectrally constrained mean-reverting diffusion process. Another direction is to redesign the residual bridge to account for the geometry of the learned mirror space. Successfully combining NAMM constraints with EMRDM-style restoration could potentially yield further improvements in both reconstruction fidelity and spectral consistency.

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